Student Weekly Lab Setup and Execution Guide

Cybersecurity Training Programme

AUCA

CENTER OF EXCELLENCE

Purpose: This guide explains the five-phase weekly workflow that all students must follow for every lab assignment in the Cybersecurity Training Programme.

Table of Contents

1. Introduction and Programme Overview

Welcome to the Cyber Defense and Response Training Programme. This programme prepares you for roles as SOC Analysts, Cyber Incident Responders, and Threat Intelligence Analysts through hands-on practical labs that simulate real-world defensive cybersecurity operations.

The programme uses a blue-team, defense-oriented approach. You will learn to:

- Identify security controls and classify them correctly
- Detect suspicious activity in logs, networks, and systems
- Respond to incidents using structured procedures
- Document findings in professional analyst-style reports
- Map observations to frameworks like NIST CSF and MITRE ATT&CK

Every week includes one lab scenario aligned to that week's module topic. You will complete the lab using virtual machines, screenshots, sample evidence, or case packs depending on your available resources.

2. Core Lab Philosophy

This programme emphasizes practical defensive skills over theoretical memorization. The labs are designed with these principles:

- **Safe and legal:** All labs use local virtual machines, instructor-provided materials, or simulated environments. You will never attack external systems.
- **Defensive focus:** You act as a defender analyzing evidence, not as an attacker breaking systems.
- **Reasoning over memorization:** Grading prioritizes your ability to explain why something matters, not just what you observed.
- **Flexible setup:** Labs support full home labs (virtual machines), partial labs (screenshots + one OS), or documentation-only routes for students with limited resources.
- **Structured documentation:** All labs require clear, concise analyst-style writing similar to what SOC teams produce in real operations.

3. Weekly Lab Workflow Overview

Every week, you will follow the same five-phase workflow. This consistency helps you build good habits and ensures you complete labs efficiently.

Phase	Activity	Time Estimate
Phase 1	Organize workspace and create weekly folder structure	5 minutes
Phase 2	Download required tools, scenarios, and materials	10-20 minutes
Phase 3	Install and configure tools in correct order	30-60 minutes (first time); 10 min (ongoing)
Phase 4	Test that environment works before starting lab	10-15 minutes
Phase 5	Perform weekly lab, collect evidence, write report	1-2 hours

Total weekly time commitment: Approximately 2-3 hours per week (including setup, lab work, and documentation).

Important: Phase 3 (installation) takes longer the first time. After Week 1, most tools remain installed, so Phases 2-3 become much faster.

4. Phase 1: Organize Your Lab Workspace

Good organization prevents confusion and file loss. Create this folder structure once at the beginning of the course, then use it consistently every week.

4.1 Main Folder Structure

Create one main folder on your computer called:

Cyber_Defense_Labs

Inside this main folder, create subfolders for each week:

- Week01
- Week02
- Week03
- ... continue through Week10

4.2 Inside Each Weekly Folder

Every week folder should contain these six subfolders:

Subfolder Name	What Goes Here
01_Instructions	Weekly scenario, lab brief, worksheet template
02_Downloads	VM images, sample logs, case packs provided by instructor
03_Installers	Tool installers (.exe, .deb, .iso files)
04_Screenshots	Evidence screenshots captured during lab
05_Evidence	Log files, notes, observations, analysis tables
06_Submission	Final worksheet, report, screenshots—ready to submit

Example complete path:

C:\Cyber_Defense_Labs\Week01\04_Screenshots\firewall_rules.png

4.3 Why This Matters

- ✓ Prevents mixing files from different weeks
- ✓ Makes it easy to find your work when writing reports
- ✓ Helps instructors grade your submission quickly
- ✓ Builds professional organizational habits used in real SOC environments

5. Phase 2: Download Required Tools and Materials

Each week, you will download only the tools and materials needed for that specific lab. Do not download everything at once—this wastes disk space and creates confusion.

5.1 What to Download Each Week

1. Weekly scenario document (provided by instructor via LMS or email)
2. Student worksheet template (provided by instructor)
3. Any case packs, sample logs, or screenshots (if provided)
4. Week-specific tools (see Section 10 for tool list by module)
5. VM images or OS installers (only needed Week 1, then reuse)

5.2 Download Checklist

Before starting the lab, verify:

✓ You saved the weekly scenario PDF/DOCX in 01_Instructions

✓ You saved the worksheet template in 01_Instructions

✓ Any tool installers are saved in 03_Installers

✓ Any instructor-provided files (logs, screenshots, case materials) are saved in 02_Downloads

✓ You noted the week number and topic clearly

5.3 Important Download Rules

- Download tools only from official sources (VirtualBox.org, Ubuntu.com, Wireshark.org, etc.)
- Verify file integrity using checksums when available
- Never download cracked or pirated software
- Check file size before downloading—some VM images are large (2-4 GB)
- Use instructor-provided links when available to ensure correct versions

6. Phase 3: Install and Configure Tools

Installation should follow a specific order to avoid problems. Install virtualization first, then operating systems, then support tools.

6.1 First-Time Installation (Week 1)

In Week 1, you will install the core lab environment that will be reused every week.

Step-by-step installation order:

- Install virtualization software: VirtualBox (free) or VMware Workstation Player (free for non-commercial use)
- Verify virtualization is enabled in BIOS/UEFI (search online for your laptop model if needed)
- Create your first virtual machine: Windows 10 or Windows 11 (use evaluation ISO or student license)
- Test the Windows VM: Boot it, verify network connectivity, take one test screenshot
- Create your second virtual machine: Ubuntu Desktop (latest LTS version)
- Test the Ubuntu VM: Boot it, verify network connectivity, open terminal
- Install Wireshark on your host computer or inside one VM
- Install documentation tools: Microsoft Word, LibreOffice Writer, or Google Docs access

6.2 Ongoing Installation (Weeks 2-10)

After Week 1, most core tools remain installed. You will only install week-specific tools as needed.

Examples:

- Week 4-5: Install Nmap for network scanning
- Week 7: Install malware analysis tools or use pre-configured VMs
- Week 9: Install Burp Suite Community Edition for web testing
- Weeks with monitoring focus: Use existing VMs and built-in tools

6.3 Installation Troubleshooting Tips

- Ensure laptop meets minimum requirements: 8GB RAM (16GB preferred), 50GB free disk space
- If VMs run slowly, reduce VM memory allocation (2GB per VM minimum)
- If virtualization fails, check BIOS settings (Intel VT-x / AMD-V must be enabled)
- Save VM snapshots after successful installation so you can restore if something breaks
- Ask instructor or lab assistant for help early—don't wait until assignment deadline

7. Phase 4: Test Your Lab Environment

Before starting the actual lab assignment, you must verify that your environment works correctly. This prevents wasting time during the lab with technical problems.

7.1 Pre-Lab Testing Checklist

Complete this checklist before every lab:

☐ Hardware check: Laptop has enough free disk space (check at least 10GB free)

☐ VM boot test: Both Windows and Ubuntu VMs boot successfully

☐ Network test: VMs can access internet (test with ping or web browser)

☐ Screenshot test: You can capture and save screenshots to 04_Screenshots folder

☐ Tool access test: Required weekly tool opens and runs without errors

☐ Worksheet access: You can open and edit the weekly worksheet document

☐ Scenario review: You have read the weekly scenario and understand the task

7.2 What to Do If Tests Fail

- VM won't boot: Restore from previous snapshot or recreate VM
- No internet in VM: Check VM network settings (NAT or Bridged mode)
- Tools don't work: Reinstall tool or check instructor-provided alternatives
- Disk space low: Delete old downloads, empty recycle bin, move files to external drive
- Performance too slow: Close background programs, reduce VM memory allocation

Important: If you cannot resolve technical issues within 30 minutes, contact the instructor immediately. Do not wait until the submission deadline.

8. Phase 5: Perform the Weekly Lab Assignment

This is the main lab work phase. You will follow the same execution pattern every week, adapted to that week's topic.

8.1 Standard Lab Execution Sequence

Follow these steps in order:

6. Read the weekly scenario completely (2-3 times if needed)
7. Identify the organization, systems, users, and security problem
8. List the assets and tools you will use
9. Open or start the required environment (VMs, screenshots, or case pack)
10. Observe or simulate the required controls or evidence
11. Capture evidence (screenshots, log excerpts, notes) and save in 04_Screenshots and 05_Evidence
12. Complete the worksheet questions with clear reasoning
13. Write the final weekly analysis note or report
14. Organize all required files in 06_Submission folder

8.2 Evidence Collection Best Practices

- Label screenshots clearly: "Week03_firewall_rules.png" not "Screenshot1.png"
- Include date/time in screenshots when possible
- Capture full window context, not just tiny cropped sections

- Write short evidence notes: "This screenshot shows Windows Defender Firewall is enabled with default rules"
- Save log files as text files with clear names: "Week05_auth_log_excerpt.txt"
- Keep evidence organized—don't have 50 unlabeled files scattered randomly

8.3 Worksheet Completion Guidelines

Most weeks include a structured worksheet with sections. Follow these guidelines:

- Answer all required questions—incomplete worksheets lose marks
- Use complete sentences—not just keywords or bullet fragments
- Justify your reasoning: Explain WHY something is a preventive control, not just that it is
- Reference evidence: "As shown in screenshot Week01_firewall.png, the firewall is enabled"
- Use technical terms correctly (review module content if unsure)
- Keep answers concise: 2-4 sentences per question is usually enough

8.4 Writing the Weekly Analysis Note

Most weeks require a final short report or analysis note (usually 300-500 words). Use this structure:

- Summary: Briefly state the organization, scenario, and main security concern (2-3 sentences)
- Observations: Describe what you found—controls, gaps, evidence (1-2 short paragraphs)
- Analysis: Explain why findings matter for confidentiality, integrity, availability, or risk (1 paragraph)
- Recommendations: Provide 3-5 prioritized, realistic actions to improve security (1 paragraph)
- Conclusion: Restate the most important finding or action (1-2 sentences)

Example opening sentence:

"AUCA Community Health Office uses a Windows workstation and Ubuntu server to store sensitive records, but the environment shows weak log monitoring and untested backup procedures."

9. Weekly Submission Requirements

Every week, you must submit the following files organized in one compressed archive or one folder.

9.1 Required Submission Files

File	Description
Completed worksheet	All questions answered clearly (DOCX or PDF)
Screenshots	Minimum 3-5 labeled screenshots showing evidence
Analysis note / report	300–500-word report (DOCX or PDF)
Evidence files (optional)	Log excerpts, tables, diagrams if required by assignment
Readme file (optional)	Short note explaining submission structure if complex

9.2 Submission Format

Package your submission as:

- LastName_FirstName_WeekXX.zip (preferred), or
- LastName_FirstName_WeekXX folder uploaded to LMS

Example: Condo_Ernest_Week01.zip

9.3 Submission Checklist Before Upload

Verify before submitting:

- ☐ All required files are included
- ☐ Files are clearly named and organized
- ☐ Worksheet is complete (no blank answers)
- ☐ Report meets word count requirement (300-500 words typical)
- ☐ Screenshots are clear and labeled
- ☐ Your name and week number appear on all documents
- ☐ File size is reasonable (under 50MB—compress large images if needed)
- ☐ You kept a backup copy on your computer

10. Recommended Tools by Module

This table shows which tools are needed for different programme modules. Install tools only when you reach that module.

Module / Week Range	Core Tools Needed	Download Source
Week 1: Foundation	VirtualBox, Windows VM, Ubuntu VM, Wireshark	virtualbox.org, microsoft.com, ubuntu.com, wireshark.org
Week 2: Threat Intel	Web browser, note-taking tool	Use existing tools
Week 3: Reconnaissance	Browser, WHOIS lookup, simple OSINT tools	Online tools (no install needed)
Week 4-5: Scanning	Nmap, Wireshark	nmap.org
Week 5: Vuln Assessment	Nessus (trial) or OpenVAS	tenable.com, openvas.org
Week 6: System Hacking	Existing VMs, password tools (instructor provided)	Use VMs from Week 1
Week 7: Malware	Isolated VM, basic	Instructor-provided

Analysis	analysis tools	safe samples
Week 8: Network Security	Wireshark, VM firewall tools	Use existing tools
Week 9: Web Security	Burp Suite Community, browser tools	portswigger.net
Week 10: Cryptography	OpenSSL, browser (for cert inspection)	Included in Ubuntu or use online tools

Note: Many tools are already included in Ubuntu or can be used via online services. Always prioritize safety and use only instructor-approved tools.

11. Troubleshooting Common Issues

11.1 Virtual Machine Problems

Problem	Solution
VM won't start	Check BIOS virtualization is enabled; reduce VM memory to 2GB; restart host computer
VM is very slow	Close background programs; allocate more RAM to VM (up to 50% of host RAM); use SSD if available
No internet in VM	Change VM network to NAT mode; check host internet works; restart VM network adapter

11.2 Tool Installation Problems

Problem	Solution
Tool won't install	Check you downloaded correct version for your OS; run installer as administrator; disable antivirus temporarily
Tool gives errors	Check you meet system requirements; search error

	message online; ask instructor for alternative tool
Can't find downloaded file	Check Downloads folder; re-download if needed; save to 03_Installers immediately

11.3 Assignment Problems

Problem	Solution
Don't understand scenario	Read it 2-3 times slowly; highlight key information; review module content; ask instructor for clarification
Can't capture evidence	Use Windows Snipping Tool or Ubuntu Screenshot app; use phone camera as backup; ask for instructor-provided screenshots
Report too short/long	Use structure in Section 8.4; focus on most important 3-5 points; remove filler sentences; add specific examples

Remember: It is always better to ask for help early than to submit incomplete or incorrect work.

12. Academic Integrity and Safe Practice

12.1 Academic Honesty Policy

All lab work must be your own individual work. You are encouraged to discuss concepts with classmates, but you must:

- ✓ Complete all lab tasks yourself
- ✓ Capture your own screenshots
- ✓ Write your own analysis and reports
- ✓ Cite any external sources you reference
- ✗ Never copy another student's screenshots or report
- ✗ Never share your completed lab files with others
- ✗ Never submit work done by someone else as your own

Violations of academic integrity will result in zero marks for the assignment and may lead to programme dismissal.

12.2 Safe and Legal Lab Practice

This programme emphasizes legal, ethical, defensive security practices. You must:

- ✓ Use only local VMs or instructor-provided lab environments
- ✓ Never scan, test, or attack external networks or systems
- ✓ Never access systems or data without explicit authorization
- ✓ Follow all instructor safety guidelines for each lab

- ✓ Report any accidental security issues immediately
- ✗ Never use programme skills for illegal purposes
- ✗ Never share exploits or attack tools outside the programme
- ✗ Never perform unauthorized security testing on AUCA or government networks

Remember: The goal is to learn defensive cybersecurity to protect organizations, not to learn hacking for harmful purposes.

13. Quick Reference Checklist

Use this checklist every week to ensure you complete all required steps.

Pre-Lab Setup (Do Once Per Week)

☐ Created this week's folder structure (Week0X with 6 subfolders)

☐ Downloaded weekly scenario and worksheet

☐ Downloaded any instructor-provided materials

☐ Downloaded and installed any new tools needed this week

☐ Tested that VMs boot correctly

☐ Tested that required tools open and work

☐ Read the weekly scenario completely

During Lab Execution

☐ Identified assets, systems, and security problems from scenario

☐ Started required environment (VMs, screenshots, or case pack)

☐ Observed and documented required evidence

☐ Captured 3-5 labeled screenshots and saved in 04_Screenshots

☐ Completed all worksheet questions with clear reasoning

☐ Wrote final analysis note (300-500 words)

☐ Checked that all answers reference evidence

Before Submission

☐ Worksheet is complete with no blank answers

☐ Screenshots are clear and properly labeled

☐ Report meets word count and follows structure

☐ All files saved in 06_Submission folder

☐ Files compressed as LastName_FirstName_WeekXX.zip

☐ Checked file size is reasonable (under 50MB)

☐ Kept backup copy on computer

☐ Submitted before deadline

Summary: The Five-Phase Weekly Cycle

Master this workflow once, and every lab becomes easier.

Phase 1: ORGANIZE	**Create weekly folder structure (6 subfolders)**
Phase 2: DOWNLOAD	Get scenario, worksheet, tools, materials
Phase 3: INSTALL	Set up tools in correct order, save snapshots
Phase 4: TEST	Verify environment works before starting lab
Phase 5: PERFORM	Execute lab, collect evidence, write report
SUBMIT	Package files as LastName_FirstName_WeekXX.zip

Questions? Contact your instructor or lab assistant immediately—don't wait until deadline day.

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